3:42 PM – 3:49 PM

Abstract citation ID: kaae014.0957

CLINICIANS AS DESIGN PARTNERS IN MHEALTH SYSTEM DEVELOPMENT

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Background: Mobile health technologists have long sought to leverage machine learning and passive sensing into instruments to detect risky patient behaviors (e.g., smoking, overeating) in vivo and report them to clinicians. Wearable cameras provide accurate detection and visual confirmation, but elements of their design preclude clinical viability. We propose a strategy for involving clinicians in the design process via focus groups to identify barriers and facilitators to adoption to inform future design iterations. We present methods and results from 2 such focus groups we conducted to improve the design of a wearable camera prototype.

Methods: 18 clinical dietitians participated in a focus group regarding a wearable camera designed to capture food intake and eating behaviors. 18 smoking cessation specialists participated in a focus group regarding a wearable camera designed to capture carbon monoxide exposure and smoking behaviors. In both groups, participants were asked probing questions of increasing specificity regarding challenges of their practice, patient metrics they would most prefer to know, and barriers they foresee in implementing our prototype. For all questions, participants were encouraged to answer as they saw fit, elaborate on their responses, and agree or disagree with statements made by fellow participants. Lastly, we surveyed participants' interest in using the device in their practice (1-5, 1=not interested, 5=extremely interested). To thematically analyze the verbal data, preliminary themes were extracted from session transcripts by independent coders who then met to compare codebooks, resolve discrepancies, and reach consensus on a final set of themes.

Results: For mean interest, cessation specialists reported 4.14/5 (n=18, SD=0.69) and dietitians reported 4.61/5 (n=18, SD=0.60). Both groups desired measures more robust than self-report, expressed interest in context factors surrounding health behaviors, and predicted implementation barriers such as operational conditions causing system failures and low adherence resulting from the device's form-factor and/or collection of image data. Some themes were unique to a group: smoking cessation specialists reported interest in the use of other psychotropic substances in conjunction with tobacco, and dietitians reported interest in the temperatures of foods consumed. All but one theme (user dishonesty) related directly to one or more technical considerations, and thus were easily incorporated into future design iterations.

Conclusion: Our survey results suggest substantial clinician interest in effective mHealth technologies. Our thematic results outline key considerations to the clinical viability of such systems. Feedback from this study significantly improved the design and optimization of our prototype, paving the way for feasibility trials of wearable cameras among treatment providers.

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Friday
March 15th, 2024
5:00 PM – 5:50 PM
Poster Session D

Abstract citation ID: kaae014.0958

POSTER SESSION D: ASSESSING NEED FOR A NETWORK-BASED PHYSICAL ACTIVITY INTERVENTION FOR CAREGIVERS OF PEOPLE LIVING WITH AD/ADRD

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Background: Unpaid family caregivers of persons living with Alzheimer's Disease (AD) and Related Dementias (AD/ADRD) often experience poor health outcomes. Because providing care often requires substantial time and other resources, caregivers may have reduced capacity to devote to their own wellbeing. However, regular physical activity is demonstrated strategy to promote both physical and mental health. Therefore, this research examined caregiver interest in an online program to leverage their social support networks and promote physical activity.

Methods: We conducted an online survey of family AD/ADRD caregivers (n= 161) related to their perceived caregiver burden and interest in an online physical activity support program. The sample was predominantly women (79.5%), white (76.4%), and had an average age of 56.39 (SD= 15.78) years. Most caregivers were either a child of (50.4%) or spouse/ partner of (38.5%) their care recipient. Participants completed the short-form Burden Scale for Family Caregivers (BSFC-SF), single-item physical and mental health ratings, and several questions related to their interest in a potential intervention.

Results: Participants indicated a strong desire to be more physically active (M= 5.83/ 7; SD= 1.19) and that they would likely be more active if they had more time (M= 4.46/ 7; SD= .69). Caregiver burden was inversely related to participants' self-reported physical (F= 15.53, p< .001, R^2 = .09) and mental (F= 35.08, p< .001, R^2 = .18) health. Caregivers in the lowest quartile of self-reported burden were significantly less interested in a support program (F= 2.83, p= .04). There were no observed differences between quartiles when examining perceived network-wide interest in participating (F= 1.73, p= .16).

Conclusions: Caregivers in our sample generally wanted to be more physically active and reported care activities as a primary barrier to doing so. Consistent with others, we found that high levels of care burden were associate with reduced physical and mental health. Further, those who reported the highest levels of care burden were most likely to indicate interest in a network-based program to provide support for them to be physically active. Participants generally felt that member of their support networks could help provide time and resources to aid in their activity. As such, there is a demonstrated need to develop interventions to promote physical activity for caregivers of people living with dementia.

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